Tuple

Tuple creation

In [1]:
```
tup1 = () # empty tuple
tup1
```

Out[1]:
```
()
```

In [8]:
```
type(tup1)
```

Out[8]:
```
tuple
```

In [7]:
```
tup2 = (10,20,30) # tuple of integers
```

In [6]:
```
tup3 = (10.44, 7.58, 6.23, 9.35) # tuple of float numbers
```

In [5]:
```
tup4 = ('one', 'two','three') # tuple of strings
```

In [4]:
```
tup5 = ('Hema', 25, (50,100), (150, 200)) # Tuple of mixed data types
```

In [3]:
```
tup6 = (1000, 'Hema', 35,87, True) #Tuple of mixed data types
```

In [9]:
```
tup7 = ('Hema', 25, [50,150],{'john','David'},(99,22,33))
```

In [10]:
```
len(tup7)
```

Out[10]:
```
5
```

Tuple indexing

In [11]:
```
tup2
```

Out[11]:
```
(10, 20, 30)
```

In [12]:
```
tup2[0] # retrieve oth index element
```

Out[12]:
```
10
```

In [13]:
```
tup4
```

Out[13]:
```
('one', 'two', 'three')
```

In [14]:
```
tup4[2] # retrieve 2nd index element
```

Out[14]: 'three'

In [15]:
```
tup6
```

Out[15]: (1000, 'Hema', 35, 87, True)

In [16]:
```
tup6
```

Out[16]: (1000, 'Hema', 35, 87, True)

In [24]:
```
tup5[0][1] # nexted indexing - access the 0th index of the 1st tuple element
```

Out[24]: 'e'

In [26]:
```
tup6[0][0] # nexted indexing - allows only on string data types
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[26], line 1
----> 1 tup6[0][0] # nexted indexing - allows only on string data types

TypeError: 'int' object is not subscriptable
```

In [27]:
```
tup6[1][0] # even with multiple data types of tuple it will retrive only string dat
```

Out[27]: 'H'

In [28]:
```
tup4
```

Out[28]: ('one', 'two', 'three')

In [29]:
```
tup4[-1]
```

Out[29]: 'three'

In [31]:
```
tup4[-1][-3] # nexted indexxing - backwrd indexing
```

Out[31]: 'r'

Tuple slicing

In [33]:
```
mytuple = ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')
```

In [34]:
```
mytuple[::]
```

Out[34]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')

In [35]:
```
mytuple[:]
```

Out[35]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')

In [36]:
```
mytuple[0:3]
```

Out[36]:
```
('one', 'two', 'three')
```

In [37]:
```
mytuple[0:6:3]
```

Out[37]:
```
('one', 'four')
```

In [41]:
```
mytuple[-1:-7:-4]
```

Out[41]:
```
('nine', 'five')
```

In [42]:
```
mytuple[-1]
```

Out[42]:
```
'nine'
```

Remove and change items in the Tuple

In [43]:
```
mytuple
```

Out[43]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')
```

In [45]:
```
del mytuple[0] # Tuples are immutable which means we can't DELETE the tuple items
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[45], line 1
----> 1 del mytuple[0] # Tuples are immutable which means we can't DELETE the tuple
items

TypeError: 'tuple' object doesn't support item deletion
```

In [46]:
```
mytuple
```

Out[46]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')
```

In [48]:
```
mytuple[0] = 1 # Tuples are immutable which means we can't chnage the value of Tupl
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[48], line 1
----> 1 mytuple[0] = 1

TypeError: 'tuple' object does not support item assignment
```

In [49]:
```
del mytuple # Deleting entire Tuple object is possible
```

In [50]:
```
type(mytuple) # bcz its already deleted
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[50], line 1
----> 1 type(mytuple)

NameError: name 'mytuple' is not defined
```

In [ ]:
```
# Loop through a tuple
```

In [51]:
```
mytuple
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[51], line 1
----> 1 mytuple

NameError: name 'mytuple' is not defined
```

In [52]:
```
mytuple = ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')
```

In [53]:
```
for i in mytuple:
    print(i)
```

```
one
two
three
four
five
six
seven
eight
nine
```

In [54]:
```
for i in enumerate(mytuple): # will return in indexed form
    print(i)
```

```
(0, 'one')
(1, 'two')
(2, 'three')
(3, 'four')
(4, 'five')
(5, 'six')
(6, 'seven')
(7, 'eight')
(8, 'nine')
```

Tuple membership

In [55]:
```
mytuple
```

Out[55]:
```
('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')
```

In [56]:
```
'one' in mytuple # check if 'one' exist in the list
```

Out[56]: True

In [57]:
```
'1' in mytuple
```

Out[57]: False

In [58]:
```
if 'three' in mytuple:
    print('three is present in the tuple')
else:
    print('three is not present in the tuple')
```

```
three is present in the tuple
```

In [59]:
```
if 'ten' in mytuple:
    print('ten is present in the tuple')
else:
    print('ten is not present in the tuple')
```

```
ten is not present in the tuple
```

index position

In [60]:
```
mytuple
```

Out[60]: ('one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine')

In [64]:
```
mytuple.index('four') # will return index of the value
```

Out[64]: 3

In [65]:
```
mytuple.index('nine')
```

Out[65]: 8

sorting

In [66]:
```
mytuple1 = (46, 76, 32, 79, 23, 12, 90, 4, 0, 167, 822)
```

In [67]:
```
sorted(mytuple1) # sorting in tuple
```

Out[67]: [0, 4, 12, 23, 32, 46, 76, 79, 90, 167, 822]

In [70]:
```
sorted(mytuple1, reverse = True) # descending order
```

Out[70]: [822, 167, 90, 79, 76, 46, 32, 23, 12, 4, 0]

In [76]:
```
mytuple1.count(79)
```

Out[76]: 1

In [77]:
```
mytuple1 * 3
```

Out[77]:
```
(46,
 76,
 32,
 79,
 23,
 12,
 90,
 4,
 0,
 167,
 822,
 46,
 76,
 32,
 79,
 23,
 12,
 90,
 4,
 0,
 167,
 822,
 46,
 76,
 32,
 79,
 23,
 12,
 90,
 4,
 0,
 167,
 822)
```

In [83]:
```
print(mytuple1)
mytuple1.count(822)
```

```
(46, 76, 32, 79, 23, 12, 90, 4, 0, 167, 822)
```

Out[83]:
```
1
```

Tuple completed